IF THE CATEGORIES OF DOTTED ARROWS ARE SETOIDS THEN THE DIAGONAL IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The second diagonal of a morphism of algebraic stacks through its inertia, and the criteria expressing separatedness of the diagonal through the closedness of that diagonal. Invoked at the start of the counted proof (source lines 9470-9477) as pointers to lemmas of this chapter. Reduces the assertion to the universal closedness of the second diagonal, which is quasi-compact by hypothesis.
- The valuative criterion for the second diagonal, involving a 2-commutative diagram from a fraction field and a choice of 2-morphism, and the description of the categories of dotted arrows. The statement refers to the relevant definition and diagram, and the proof uses the corresponding lemma of this chapter (source lines 9480-9511). Supplies the shape of the diagrams to be filled and the automorphisms to be compared.

Source reference. Original definition definition-fill-in-diagram: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9116–9116. Navigation pointer only: the counted text refers to the Stacks item definition-fill-in-diagram, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original equation equation-diagram: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9080–9080. Navigation pointer only: the counted text refers to the Stacks item equation-diagram, which belongs to another chapter of the project and is not part of this card. The pointer

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Source reference. Original lemma lemma-base-change-existence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9514–9514. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-existence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-cat-dotted-arrows: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9447–9447. Navigation pointer only: the counted text refers to the Stacks item lemma-cat-dotted-arrows, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-diagonal-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3959–3959. Navigation pointer only: the counted text refers to the Stacks item lemma-diagonal-diagonal, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-first-diagonal-separated-second-diagonal-closed: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1816–1816. Navigation pointer only: the counted text refers to the Stacks item lemma-first-diagonal-separated-second-diagonal-closed, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-quasi-compact-existence-universally-closed: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9527–9527. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-quasi-compact-existence-universally-closed, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. If Δ_f is quasi-separated and if for every diagram (equation-diagram) and choice of γ as in Definition definition-fill-in-diagram the category of dotted arrows is a setoid, then Δ_f is separated.*

Proof. We are going to write out a detailed proof, but we strongly urge the reader to find their own proof, inspired by reading the argument given in the proof of Lemma lemma-cat-dotted-arrows.

Assume Δ_f is quasi-separated and for every diagram (equation-diagram) and choice of γ as in Definition definition-fill-in-diagram the category of dotted arrows is a setoid. By Lemma lemma-diagonal-diagonal it suffices to show that $e : \mathcal{X} \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is a closed immersion. By Lemma lemma-first-diagonal-separated-second-diagonal-closed it in fact suffices to show that $e = \Delta_{f,2}$ is universally closed. Either of these lemmas tells us that $e = \Delta_{f,2}$ is quasi-compact by our assumption that Δ_f is quasi-separated.

In this paragraph we will show that e satisfies the existence part of the valuative criterion. Consider a 2-commutative solid diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle e} \\ \mathrm{Spec}(A) & \xrightarrow{(a,\theta)} & \mathcal{I}_{\mathcal{X}/\mathcal{Y}} \end{array}$$

and let $\alpha : (a, \theta) \circ j \to e \circ x$ be any 2-morphism witnessing the 2-commutativity of the diagram (we use α instead of the letter γ used in Definition definition-fill-in-diagram). Note that $f \circ \theta = \mathrm{id}$; we will use this below. Observe that $e \circ x = (x, \mathrm{id}_x)$ and $(a, \theta) \circ j = (a \circ j, \theta \star \mathrm{id}_j)$. Thus we see that α is a 2-arrow $\alpha : a \circ j \to x$ compatible with $\theta \star \mathrm{id}_j$ and id_x. Set $y = f \circ x$ and $\beta = \mathrm{id}_{f \circ a}$. Reading the arguments given in the proof of Lemma lemma-cat-dotted-arrows backwards, we see that θ is an automorphism of the dotted arrow (a, α, β) with

$$\gamma : y \circ j \to f \circ x \quad \text{equal to} \quad \mathrm{id}_f \star \alpha : f \circ a \circ j \to f \circ x$$

On the other hand, id_a is an automorphism too, hence we conclude $\theta = \mathrm{id}_a$ from the assumption on f. Then we can take as dotted arrow for the displayed diagram above the morphism $a : \mathrm{Spec}(A) \to \mathcal{X}$ with 2-morphisms $(a, \mathrm{id}_a) \circ j \to (x, \mathrm{id}_x)$ given by α and $(a, \theta) \to e \circ a$ given by id_a.

By Lemma lemma-base-change-existence any base change of e satisfies the existence part of the valuative criterion. Since e is representable by algebraic spaces, it suffices to show that e is universally closed after a base change by a morphism $I \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ which is surjective and smooth and with I an algebraic space (see Properties of Stacks, Section stacks-properties-section-properties-morphisms). This base change $e' : X' \to I'$ is a quasi-compact morphism of algebraic spaces which satisfies the existence part of the valuative criterion and hence is universally closed by Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-existence-universally-closed. $\square$